Optimization Strategy for Ultra-short-term Photovoltaic Power Forecasting based on the Fusion of Day-ahead Forecast and Intraday Real-time Information

Cheng Jifeng
Electric Power Science Research State Grid Liaoning Eleetric Power Co., Ltd
Shenyang 110000, China
cjf ldk@163.com

Sun Junjie
Electric Power Science Research State Grid Liaoning Eleetric Power Co., Ltd
Shenyang 110000, China
1256950778@qq.com

Wang Li
State Grid Chaoyang Power Supply Company
Chaoyang 122000, China
5880954@qq.com

Wu Zhenghao
Department of Electrical Engineering North China Electric Power University
Baoding 071003, China
zhenghaowu@ncepu.edu.cn

Wang Yuqing
Department of Electrical Engineering North China Electric Power University
Baoding 071003, China
yuqingwang@ncepu.edu.cn

Wang Fei
Department of Electrical Engineering North China Electric Power University
Baoding 071003, China
feiwang@ncepu.edu.cn

2025 IEEE Industry Applications Society Annual Meeting (IAS) | 978-1-6654-5776-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/IAS62731.2025.11061420

***Abstract*—High-precision photovoltaic ultra-short-term power forecasting holds significant theoretical importance and practical value in enhancing the stability and reliability of power systems under photovoltaic grid connection conditions. However, the traditional power extrapolation-based ultra-short-term forecasting model experiences a significant decline in forecasting performance under transitional weather conditions, as the correlation between the power sequence of the forecast period and the input period of the model is reduced. Therefore, this study proposes a novel optimization method integrating day-ahead forecasts and intraday real-time data. First, establish a long short-term memory neural network forecasting model with a two-stage architecture using historical meteorological power data. Secondly, historical similar days are determined through real-time intraday measurements, and the actual measured and day-ahead forecast power information of these similar days are obtained. This information is used to correct the errors in the day-ahead forecast of the forecasting period, in order to reduce the inherent bias of the day-ahead forecast for specific periods during the forecasting period. Thus, the corrected day-ahead forecast data is integrated into the modeling framework of ultra-short-term power forecasting. Case studies at a large-scale photovoltaic plant in Inner Mongolia demonstrate the method's effectiveness and superiority.**

Keywords—ultra-short term power forecasting, intraday real-time information fusion, weather transition, similar-day clustering

I. Introduction

Photovoltaic ultra-short-term power forecasting plays a crucial role in high proportion photovoltaic power systems. Its core significance lies in the following: Firstly, by accurately forecasting the fluctuations of photovoltaic output, it can provide forward-looking decision-making basis for power system dispatching, effectively alleviating the impact of the randomness of new energy on the stability of the power grid[1]; moreover, precise ultra-short-term forecasting helps to improve the accuracy of the connection between day-ahead and real-time transactions in the power market[2], reducing the risk of deviation assessment. As the penetration rate of photovoltaic power increases, this technology has become a key technical support for ensuring system frequency stability and preventing power shortages, directly affecting the safety and economic operation level of the power grid[3].

The main methods for ultra-short-term photovoltaic power forecasting include two types: statistical learning method and artificial intelligence method. The statistical learning method uses time series analysis, such as the Auto-regressive Integrated Moving Average model (ARIMA)[4], and is suitable for mining the regularity of historical data. The artificial intelligence method employs machine learning techniques, including Support Vector Machine (SVM)[5], Random Forest (RF)[6], and deep learning methods such as Long Short-Term Memory Network (LSTM)[7] ， Convolutional Neural Network (CNN)[8], which can efficiently capture the nonlinear features of data.

However, this type of time series extrapolation-based forecasting method based on historical power data performs well when the meteorological conditions are stable, but it is difficult to effectively address the power correlation issues caused by transitional weather[9]. When there is a daytime transitional weather, the forecasting sequence and the historical sequence are under different weather conditions, so there are certain differences in the fluctuation characteristics and amplitudes of the meteorological and power information between the two, which makes the traditional forecasting method unable to accurately capture the power evolution trend. The day-ahead power forecasting is based on NWP information and is not affected by the poor correlation between the power sequences before and after the daytime transitional weather in the forecasting modeling**Error! Reference source not found.**. Therefore, the day-ahead power forecasting information can be introduced into the power forecasting modeling to optimize the ultra-short-term forecasting.

As the optimization method based on the current day-ahead forecast power is limited by the inherent error characteristics of NWP and is difficult to be directly applied to optimize the ultra-short-term power forecasting in this scenario[10]. In contrast, the real-time daily power and meteorological data have a minute-level time resolution and can effectively represent the changing characteristics of power meteorological data in real time. Therefore, constructing a multi-time-scale fusion mechanism for the integration of day-ahead forecast information and real-time daily information can effectively compensate for the lag in model updates and the insufficient feature capture ability of the NWP model in long-term forecasting, thereby significantly improving the optimization effect of ultra-short-term power forecasting.

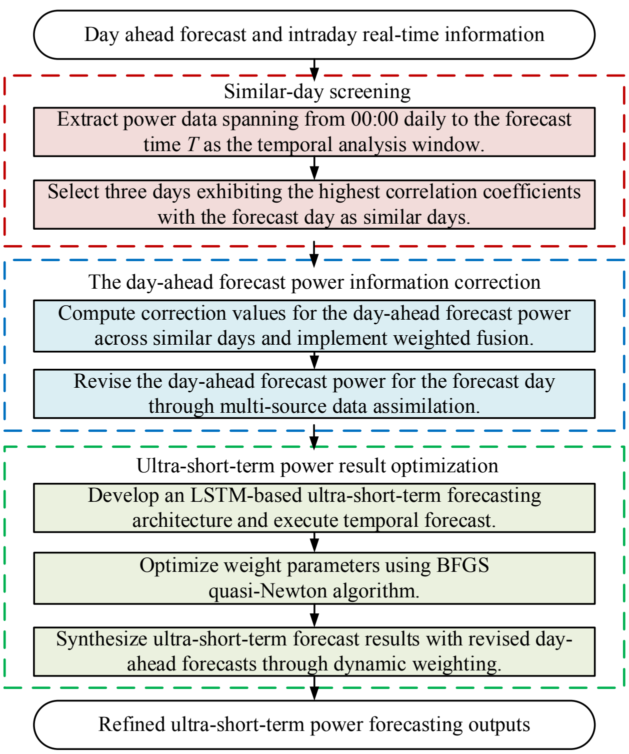

Fig. 1. Flowchart of Power Forecasting Methodology.

Based on the above analysis, this paper proposes a super-short-term power forecasting method based on the fusion of day-ahead forecast and real-time daily information as shown in Fig. 1:

1) Based on the real-time measured power, similar days are matched to obtain the historical measured power and the corresponding day-ahead forecast power information, thereby compensating for the inherent forecasting errors of the corresponding day-ahead forecast information in the forecasting period.

2) The corrected day-ahead forecast information is incorporated into the optimization modeling of ultra-short-term power forecasting, thereby effectively improving the accuracy of ultra-short-term power forecasting through the complementary effect of day-ahead forecast and real-time measured information.

The structure of the remaining part of this article is as follows: The second part presents an optimization model for ultra-short-term electricity forecasting that combines day-ahead forecastings and real-time data during the intraday period. The third part concludes the performance of this model in practical cases. The fourth part draws the conclusion.

II. Methodology

The optimization strategy described in this paper is mainly divided into three parts: the similar-day screening, the day-ahead forecast power information correction, and the ultra-short-term power result optimization.

A. Similar-day screening

To avoid the problem of weakened power correlation caused by sudden weather changes, this paper uses the Pearson correlation coefficient as the basis for selecting similar days. The Pearson coefficient can effectively capture the fluctuation characteristics and trend similarity of power sequences. This method quantifies the linear correlation degree between known power sequences within a day, and selects the historical day with the highest correlation to the power curve of the day to be forecasted as a similar day.

First, calculate the Pearson correlation coefficient of the meteorological power data $r_d$within the time period from 0:00 of each day within the previous seven days up to the forecasting time T for the forecasting day. Suppose the forecasting date is day d, the Pearson correlation coefficient on day d can be calculated according to Eq.(1).

$$r_d = \frac{\sum_{t=1}^{n}(P_{d,t}-\overline{P}_d)(P_t-\overline{P})}{\sqrt{\sum_{t=1}^{n}(P_{d,t}-\overline{P}_d)^2\sum_{t=1}^{n}(P_t-\overline{P})^2}}, (t \in [0,T],\ \ d = 1,2,\ldots,7) \qquad (1)$$

P_t and $\overline{P}$ denote the measured power and the average value at time t on the forecast day; $P_{d,t}$ and $\overline{P}_d$ represent the measured power and the average value at at time t on day d; n is the number of samples within the 0-T period, with a time resolution of 15 minutes.

In order to reduce the accidental errors caused by a single similar day, the three days with the highest correlation coefficient are selected as the similar days.

B. The day-ahead forecast power information correction

Calculate the correction value of the actual power and the day-ahead forecast power in the forecasting period for each similar day, and fuse the correction values obtained from the three similar days in a weighted manner, which can be obtained by Eq. (2) and Eq. (3).

$$\Delta\alpha_{i,j} = P_{i,j} - P_{st\,i,j}, (i \in [T, T+16],\ \ j = 1,2,3) \tag{2}$$

$$\Delta\alpha_i = 0.5\Delta\alpha_{i,1} + 0.3\Delta\alpha_{i,2} + 0.2\Delta\alpha_{i,3} \tag{3}$$

In the formula, $\Delta\alpha_i$ represents the corrected value at time *i* after weighted fusion; *j* represents the ranking of correlation for similar days, with a higher *j* value indicating a higher correlation coefficient for the corresponding similar day; $P_{i,j}$ represents the measured power value at time *i* within similar day *j*; $P_{st\,i,j}$ represents the day-ahead forecast power value at time *i* within similar day *j*; $\Delta\alpha_{i,j}$ represents the corrected value at time *i* within similar day *j*.

After calculating the corrected value $\Delta\alpha_i$ of the day-ahead forecast power through similar-day fusion, it is added to the day-ahead forecast power $P'_{st\,i}$ of the forecasting day to obtain the revised day-ahead forecast power $P''_{st\,i}$ by Eq.(4).

$$P''_{st\,i} = P'_{st\,i} + \Delta\alpha_i \tag{4}$$

C. Ultra-short-term power result optimization

The ultra-short-term power forecasting model adopted in this paper is a two-stage architecture prediction model: In the first stage, a parallel LSTM and backpropagation (BP) architecture is employed to handle heterogeneous data. The LSTM model extracts the characteristics of power data from historical periods, while the BP model extracts the characteristics of NWP data for the forecasting period. In the second stage, the extracted features are integrated through the BP network to generate ultra-short-term forecastings, achieving the collaborative utilization of multiple features.

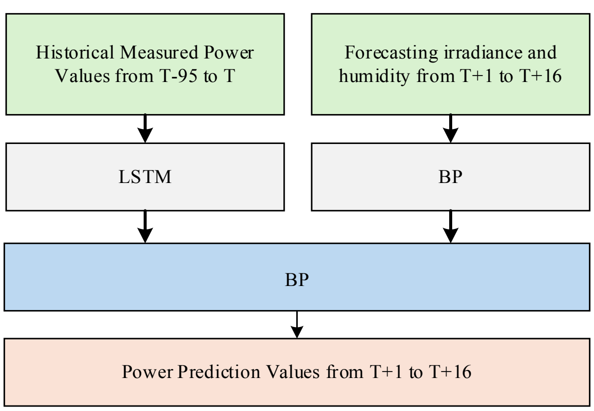

Fig. 2. Schematic diagram of the two-stage architecture prediction model.

The function of this hybrid architecture lies in the following: On the one hand, it avoids the limitations of a single model when dealing with spatio-temporal heterogeneous data. LSTM alleviates the deficiency of traditional BP networks in time series modeling, while BP networks make up for the shortcomings of LSTM in static feature learning.

On the other hand, feature-level fusion enhances the model's ability to collaboratively utilize multi-source data (historical power and meteorological forecasts), and improves the stability of the forecasting through the cross-validation mechanism.

Finally, through the BP output layer, end-to-end forecasting is achieved, enabling spatio-temporal features to jointly act on the power forecasting results. In the ultra-short-term scenario, it balances the agility of the response to meteorological sudden changes and the smoothness of the power curve, meeting the dual requirements of both.

After obtaining the ultra-short-term power forecasting results using the aforementioned forecasting model, the ultra-short-term power forecasting results are optimized by using the corrected day-ahead forecast power. The new ultra-short-term power forecasting results are obtained through weighted fusion by Eq.(5).

$$P_{cmb\,i} = (1 - w_i) \cdot P''_{st\,i} + w_i \cdot P_{ust\,i}, (w_i \in [0,1]) \tag{5}$$

$P_{ust\,i}$ and $P_{cmb\,i}$ represent the ultra-short-term power forecasting results before and after optimization at time *i* respectively, w_i and signifies the combination weight at time *i*.

This paper uses the Broyden-Fletcher-Goldfarb-Shanno (BFGS) method to conduct parameter optimization calculations for the combination weights. BFGS is a quasi-Newton optimization algorithm, which is used to solve unconstrained nonlinear optimization problems. Its core idea is to approximate the Hessian matrix or its inverse of the objective function through iterative methods, thereby avoiding the high computational cost of directly calculating the second derivatives[12].

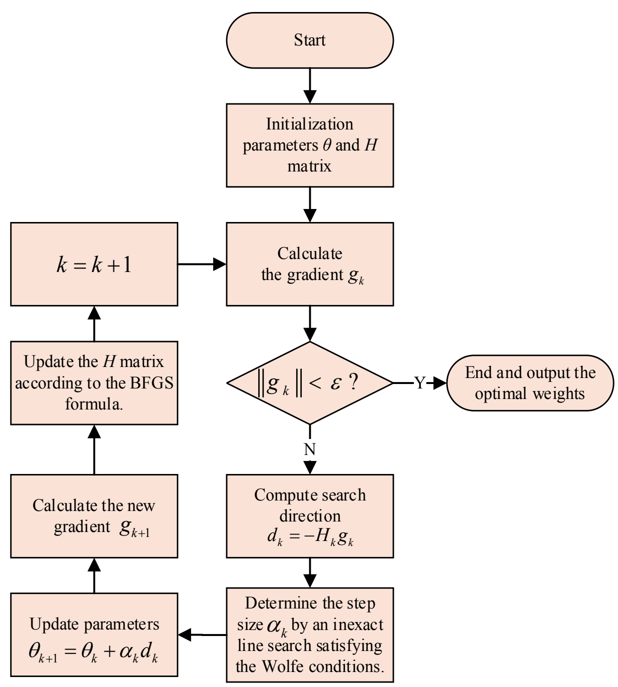

Fig. 3. Flowchart of the BFGS Iterative Process.

In this case, BFGS is used to calculate the fusion weights for ultra-short-term and short-term forecasting results, mainly based on the following advantages: Firstly, BFGS has an ultra-linear convergence speed[13], which enables efficient solution of the weight optimization problem and meets the real-time requirements of power forecasting; Secondly, BFGS only requires one gradient information, avoiding the complexity of Hessian matrix calculation, and is suitable for high-dimensional weight optimization; Compared with traditional fixed weights or simple linear weighting, BFGS can more accurately balance the sensitivity of ultra-short-term forecasting and the stability of short-term forecasting, improving the overall forecasting accuracy.

The complete BFGS iterative flowchart is shown in the figure 3. To satisfy the weight constraints, an unconstrained intermediate variable θ_i is first introduced, which is mapped to the weights w_i through the Sigmoid function, as shown in Eq.(6).

$$w_i = \sigma(\theta_i) = \frac{1}{1+e^{-\theta_i}}, (\theta_i \in \mathbb{R}) \tag{6}$$

Suppose there are *n* sub-samples in the training set N. Then the objective function of the iteration, that is, the mean square error of the entire training set, is given by Eq.(7). $P_{cmb,\,i}^{(n)}$ represents the optimized ultra-short-term forecasting power value at time *i* in the *n*-th sample, and $P_{true,\,i}^{(n)}$ represents the measured power value at time *i* in the *n*-th sample.

$$J = \frac{1}{16N} \sum_{n=1}^{N} \sum_{i=1}^{16} \left(P_{ust,\,i}^{(n)} - P_{cmb,\,i}^{(n)} \right)^2 \tag{7}$$

As shown in Eq(8), the gradient is obtained through the chain rule.

$$\frac{\partial J}{\partial \theta_i} = \frac{\sum_{n=1}^{N} \left(P_{cmb,\,i}^{(n)} - P_{true,\,i}^{(n)} \right) \left(P_{ust,\,i}^{(n)} - P_{st,\,i}^{(n)} \right) (1-w_i) \cdot w_i}{8N} \tag{8}$$

Finally, the variable θ are iterated. Firstly, the weights, *H* matrix and step size are initialized as shown in Eq.(9).

$$\theta_0 = [0,0,\cdots,0], H_0 = I, \alpha_k = 1 \tag{9}$$

Then, the iteration θ is carried out, where *k* represents the iteration number, and $\alpha_k$represents the iteration step size, which is determined by an inexact line search satisfying the Wolfe conditions[14]. $s_k$and y_k respectively represent the parameter update amount and the gradient change amount, used to update the iteration matrix *H*.

Finally, when g_k is less than the threshold, the iteration stops and the optimal weights w_i^* are output, as shown in Eq.(10).

$$\begin{cases} g_k = \nabla J(\theta_k) \\ d_k = -H_k g_k \\ \theta_{k+1} = \theta_k + \alpha_k d_k \\ s_k = \theta_{k+1} - \theta_k \\ y_k = g_{k+1} - g_k \end{cases} \tag{10}$$

III. Case Study

A. Description of Dataset

This paper uses a power generation dataset from a large-scale photovoltaic (PV) base in Inner Mongolia. The dataset contains 15-minute time-resolved photovoltaic power generation data of the 4th station from September 2022 to December 2024. The installed capacity of the 4th station is 75MW.

Among them, the period from September 1, 2022 to January 1, 2024 is the training set of the model, the period from January 1, 2024 to June 30, 2024 is the validation set of the model, and the period from July 1, 2024 to December 31, 2024 is the validation set of the model.

In the training and validation datasets of the model, abnormal power values marked as power curtailment moments are replaced with the maximum power generation capacity of the station at the same time. In the test set of the model, abnormal moments marked as power curtailment will not be included in the accuracy range.

In this paper, one commonly used metric is adopted to quantify the forecasting performance of the proposed method, which is the mean harmonic precision**Error! Reference source not found.**.

$$ACC_t = 1 - \frac{1}{Cap} \sqrt{\sum_{i=1}^{16} \left[\left(P_i - P_i' \right)^2 \cdot \frac{|P_i - P_i'|}{\sum_{i=1}^{16} |P_i - P_i'|} \right]} \tag{11}$$

$$ACC = \frac{1}{T} \sum_{t=1}^{T} ACC_t \tag{12}$$

In Eq.(11) and Eq.(12), ACC_t represents the harmonic accuracy at time *t*; ACC denotes the average harmonic accuracy over the forecasting period; P_i' and P_i are the forecasting and actual values, respectively, at the *i*-th step of the ultra-short-term forecast at time *t*; Cap signifies the maximum capacity of the photovoltaic plant, and *T* represents the forecast period.

To validate the superiority of the proposed optimization strategy, this study establishes two additional methods for comparison:

Method 1: An ultra-short-term power forecasting model based on meteorological data and historical power data.

Method 2: Building upon Method 1, the day-ahead forecast power data is weighted and fused with the ultra-short-term power forecasting results to obtain optimized ultra-short-term forecasts.

Method 3: The proposed method in this study.

TABLE I. Comparison of Different Methods

Month	Method		
	M1	*M2*	*M3*
Overall	94.03%	93.22%	**94.32%**

Table I shows the average accuracy of these three methods over a period of four months, with the best-performing method represented in bold. Fig.4 is a comparison chart of the monthly average accuracy of these three methods. From the comparison of the forecasting accuracy of the three methods, it can be seen that Method 3 demonstrates a significant overall advantage. Specifically, its forecasting accuracy has consistently led from

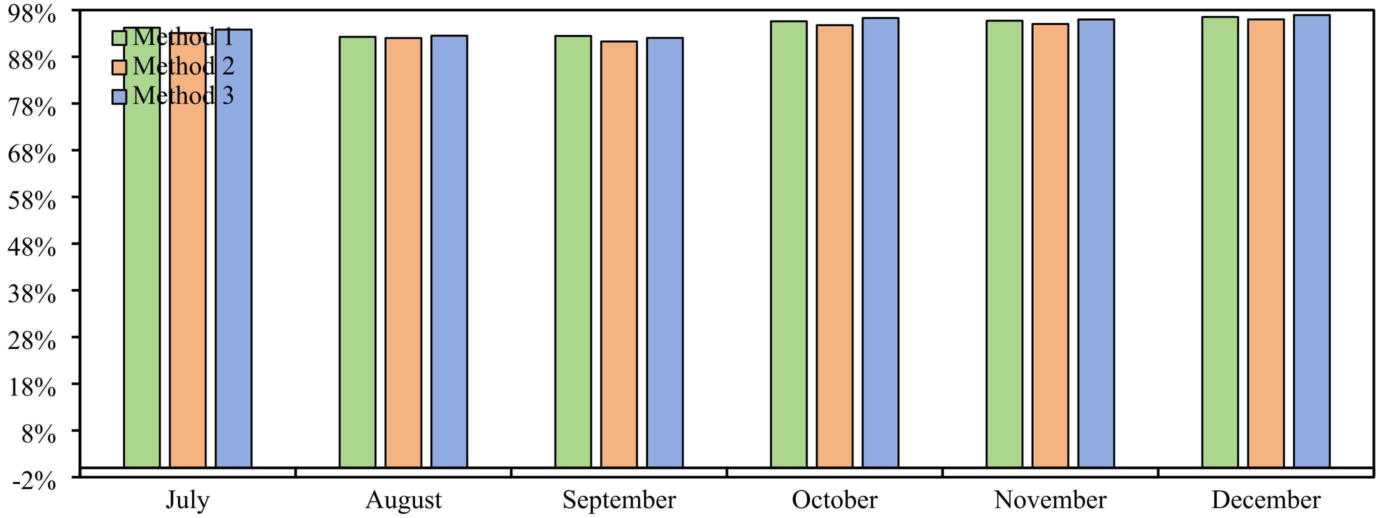

Fig. 4. Accuracy Comparison Chart of Methods (Monthly Average).

October to December, and the overall average accuracy reaches 94.33%, significantly higher than that of the other methods. This result indicates that by utilizing intraday power information and optimizing the ultra-short-term forecasting of power changes and improve the forecasting accuracy.

Select the dates when significant weather changes occurred in the test set, and calculate the accuracy of the methods 1 and 3 on those days. The comparison results of the accuracy are shown in Fig.5.

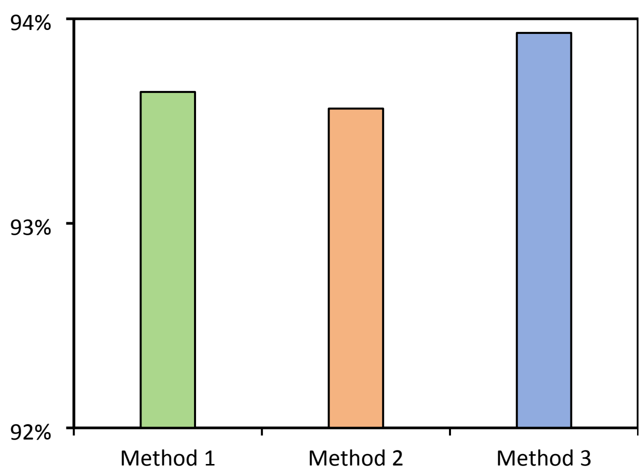

Fig. 5. Accuracy Comparison Chart of Different Methods in Transitional Weather Scenarios.

The comparison of the average accuracy of the three ultra-short-term power forecasting methods under the transitional weather scenarios reveals that Method Three has the best forecasting performance, with its accuracy slightly higher than that of Method One and Method Two. This result indicates that Method Three, by combining the strategy of optimizing the ultra-short-term results, can more effectively address the challenges posed by weather type changes to power forecasting.

When the weather type of the historical power data input by the model differs from the actual weather type of the forecasting period, Method Three can fully utilize the known power information within the day, match similar days, and obtain effective forward-looking power information to optimize the ultra-short-term forecasting results, thereby reducing the impact caused by the low correlation between the input power sequence and the power of the forecasting period.

IV. Conclusion

This paper proposes an optimization strategy for ultra-short-term photovoltaic power forecasting based on the fusion of daily-ahead forecast and real-time information during the day. This method achieves the optimization of ultra-short-term photovoltaic power forecasting results by utilizing the similarity-day screening mechanism of real-time information during the day and the BFGS dynamic weight optimization algorithm. Firstly, the correlation coefficient is calculated based on the daily information to screen historical similar days, obtain their actual power and the error of the daily-ahead forecast, and then correct the current daily-ahead forecast value; Secondly, the spatial-temporal features are extracted through feature-level fusion of LSTM and BP networks, and the BFGS algorithm is used to dynamically optimize the fusion weights of ultra-short-term forecasting and correction of the daily-ahead forecast to balance the forecasting sensitivity and stability.

Case studies show that the proposed method performs excellently on a large-scale photovoltaic power station dataset in Inner Mongolia, with an average accuracy of 94.32%, superior to traditional methods, especially demonstrating stronger robustness in transitional weather scenarios. Experimental results indicate that this ultra-short-term photovoltaic power forecasting optimization strategy effectively alleviates the problem of insufficient correlation between the input sequence and the power of the forecasting period, and improves the adaptability of the model.

Future research can be improved in the following aspects: Firstly, the generalization performance of the method in different climate regions and larger data sets needs to be verified; Secondly, a more refined weather classification model can be explored to improve the accuracy of similarity-day screening.

ACKNOWLEDGMENT

This work was supported by Science and technology project of State Grid Liaoning Electric Power Co., Ltd. (2024YF-103).

REFERENCES

[1] M. Shi, R. Yin, P. Zhang, J. Tian and Y. Wang, "Ultra short-term forecast of photovoltaic generation based on time-division LSTM neural networks," 2020 12th IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC), Nanjing, China, 2020, pp. 1-5.

[2] J. Wang, Y. Yu, B. Zeng, and H. Lu, "Hybrid ultra-short-term PV power forecasting system for deterministic forecasting and uncertainty analysis," *Energy*, vol. 288, p. 129898, Feb. 2024.

[3] Z. Su, S. Gu, J. Wang, and P. D. Lund, "Improving ultra-short-term photovoltaic power forecasting using advanced deep-learning approach," *Measurement*, vol. 239, p. 115405, Jan. 2025.

[4] J. Lee, H. Lee, N. Kim and B. Lee, "A Study on Online ARIMA Algorithms applying various gradient descent optimization algorithms for Time Series Prediction," 2021 International Conference on Information and Communication Technology Convergence (ICTC), Jeju Island, Korea, Republic of, 2021, pp. 1104-1106.

[5] R. Nguyen, Y. Yang, A. Tohmeh and H. -G. Yeh, "Predicting PV Power Generation using SVM Regression," 2021 IEEE Green Energy and Smart Systems Conference (IGESSC), Long Beach, CA, USA, 2021, pp. 1-5.

[6] G. Xie and Z. Lin, "RSMD-RF-BGSkip Based PV Generation Prediction Method," *IEEE Access*, vol. 12, pp. 65799-65809, 2024.

[7] X. Jin, R. Han and Y. Ma, "PV Power Forecasting Based on VMD-RIME-LSTM," 2024 IEEE China International Youth Conference on Electrical Engineering (CIYCEE), Wuhan, China, 2024, pp. 1-5.

[8] J. Yan et al., "Frequency-Domain Decomposition and Deep Learning Based Solar PV Power Ultra-Short-Term Forecasting Model," *IEEE Transactions on Industry Applications*, vol. 57, no. 4, pp. 3282-3295, July-Aug. 2021.

[9] G. Z. Yu, L. Lu, B. Tang, S. Y. Wang, and C. Y. Chung, "Ultra-Short-Term Wind Power Subsection Forecasting Method Based on Extreme Weather," *IEEE Transactions on Power Systems*, vol. 38, no. 6, pp. 5045-5056, Nov. 2023.

[10] E. Sarmas, E. Spiliotis, E. Stamatopoulos, V. Marinakis, and H. Doukas, "Short-term photovoltaic power forecasting using meta-learning and numerical weather prediction independent Long Short-Term Memory models," Renewable Energy, vol. 216, p. 118997, Nov. 2023.

[11] H. Dai, Z. Zhen, F. Wang, Y. Lin, F. Xu, and N. Duić, "A short-term PV power forecasting method based on weather type credibility prediction and multi-model dynamic combination," *Energy Conversion and Management*, vol. 326, p. 119501, Feb. 2025.

[12] F. Zocco and S. McLoone, "An Adaptive Memory Multi-Batch L-BFGS Algorithm for Neural Network Training," IFAC-PapersOnLine, vol. 53, no. 2, pp. 8199-8204, Jan. 2020

[13] Y. Gao, G. Wang, C. Li, M. Li, and X. Cheng, "Energy-efficient power allocation algorithm based on the BFGS algorithm and the armijo criterion," *Sustainable Energy Technologies and Assessments*, vol. 57, p. 103217, Jun. 2023.

[14] H. Chen, Q. Liu, D. Sun, S. Yu, and H. Chen, "Fast Model Predictive Control Trajectory Tracking for Autonomous Vehicles Based on BFGS Interior Point Method," IFAC-PapersOnLine, vol. 58, no. 29, pp. 433-438, Jan. 2024.

[15] B. Liu, Z. Liu, Y. Wang, Z. Zhen, F. Wang and Z. Tong, "Ultra-Short-Term Power Forecasting for Distributed PV based on Multi-Source Remote Sensing Information and Adaptive Feature Extraction," 2024 IEEE/IAS Industrial and Commercial Power System Asia (I&CPS Asia), Pattaya, Thailand, 2024, pp. 484-489.